# Parameter recovery: acuity selection on the tested grid

# Parameter recovery: acuity selection on the tested grid I

Parameter recovery probes whether the executed observation model contains enough information to distinguish sensor acuity under the study design. We sweep acuity values 0.60, 0.70, 0.80, 0.90: for each true acuity the model generates 200 synthetic observations per trial across 960 independent trials, fits acuity by marginal-likelihood grid search, and compares the recovered value with ground truth.

Across the sweep the mean absolute recovery error is 0.0232 and the coefficient of determination of mean-recovered versus true acuity is $R^2 = 0.9999$. Within this finite grid and observation budget, recovered acuity tracks the identity line with the reported error (fig. 1). This is evidence of practical acuity recoverability for the executed design, not a proof of global or structural identifiability and not an acuity-by-colony-size study.

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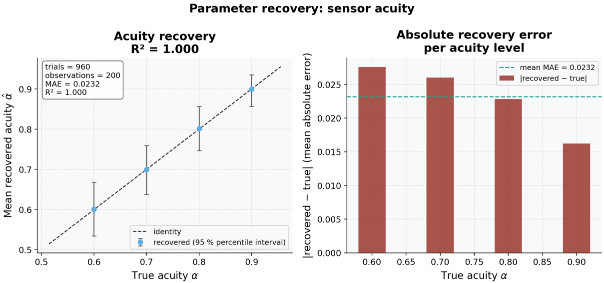


Figure 1: Two-panel parameter-recovery figure. Source relation: original project parameter-recovery diagnostic; estimand: recovered acuity and absolute error in probability units; uncertainty: empirical percentile intervals across independent trials. In the left panel, the x-axis is true acuity and the y-axis is recovered acuity, both in probability units; error bars show the 95% empirical percentile interval across 960 trials per condition, and the diagonal is the identity reference. This interval is a descriptive quantile of the independent-trial estimates, not a bootstrap confidence interval or Bayesian credible interval. In the right panel, the x-axis is tested true acuity and the y-axis is absolute acuity error in probability units; the horizontal line is the global mean absolute error. These finite-grid results quantify acuity recovery for 200 observations per trial; they do not establish global structural identifiability.